

Aiden Wenzel

(248)-775-9622 aidenwen@umich.edu linkedin.com/in/aiden-wenzel github.com/aiden-wenzel

EDUCATION

University of Michigan *BSE, Electrical Engineering*

September 2023 - May 2027

- Computer Science Minor
- GPA: 3.54/4.00

SKILLS

- **Programming Languages:** C, C++, Python, Lua, MATLAB, Verilog HDL, $\text{\LaTeX}$.
- **Linux:** Debian, WSL2, Grep, CMake, Make, Tmux, Vim, SSH.
- **Software:** Simulink, Waveforms Oscilloscope, LTSpice, ModelSim, Jupyter Notebooks.

PROFESSIONAL EXPERIENCE

Coretek Services *Technical Support Specialist*

May 2024 - June 2025

- Worked with HTML, CSS, and C# to develop intuitive ASP.NET UIs.
- Used C# APIs to pull cloud resource information from MS Azure and ticket information from ServiceNow.
- Used KQL to query and summarize cloud resource analytics in Azure dashboards.

MRI Reconstruction Lab *Student*

Dec 2026 - Present

- Worked with a team from ECE and Radiology researching reconstruction algorithms for accelerated MRI.
- Reproduced reconstruction results with Python packages like Scipy, PyTorch, and MIRTorch.
- Visually and quantitatively compared results between reconstruction methods.
- Presented my findings at the 2026 ECE undergraduate research symposium.
- Implemented optimization algorithm to approximate the B1 magnetic field maps inside MRI scanner with Julia.

ENGR 101 Instructor *Instructional Aide*

Sep 2026 - Present

- Taught introductory programming lab to incoming engineering students.
- Answered student questions through in-person office hours and forum styled posts.

PROJECTS

Image Compression Pipeline

- Collaborated with a team of 4 engineering students to create an image compression pipeline in Python.
- Utilized the Discrete Cosine Transform, Singular Value Decomposition, and Huffman Encoding to compress images.

Fractal Visualizer

- Utilized OpenGL APIs to accurately visualize the Mandelbrot set and Julia sets.
- Implemented zoom functionality for near-infinite zooming into complex geometries.

IR Room Occupancy Counter

- Utilized the MLX90640 IR array sensor and ESP32 to track the movement of heat signatures in and out of a room.
- Worked extensively with the MLX90640 API, contour tracing algorithms, and Wi-Fi communication.
- Designed custom PCB to interface MLX with the ESP32.

Instrument Recognition Software

- Worked with a team of 3 other engineering students to build an instrument recognition app.
- Input audio data would be processed using the Librosa library. Harmonic overtones in audio samples would be extracted using the FFT algorithm.
- Processed audio data would be used to train a CNN which would take audio files as input, process them, and predict what musical instrument is featured in the sample.

RELEVANT COURSE WORK

- **Digital Signal Processing:** Studied mathematical concepts such as LTI systems, convolution, FFT, and the Z-transform in the discrete domain. Utilized languages like MATLAB and Python to quickly perform calculations on vectors and matrices. Explored the applications of signal processing in communications, music, image processing, and medicine.
- **Mathematics for Machine Learning:** Utilized linear algebra and multi-variable calculus to derive important machine learning equations. Homeworks involved using Python to implement algorithms learned in class.

EXTRACURRICULARS

Michigan Marching Band *Piccolo*

August 2023 - December 2026

- Dedicated approximately 20 hours each week to rehearsals and performances in the Big House.
- Performed at the 2023 Big Ten Championship, 2024 ReliQuest Bowl, and 2025 Citrus Bowl.
- Initiated as an official brother of Kappa Kappa Psi ($\text{\textit{KK}\Psi}$), the National Honorary Band Fraternity.
- Performed with the Michigan Hockey Band at Yost Ice Arena.